

Asmi Gujral

Open to Relocating | [LinkedIn](#) | [GitHub](#) | [Portfolio Website](#) | 416-579-2023 | asmi.gujral@gmail.com

EDUCATION

University of Waterloo – Honors Mechatronics Engineering, Bachelor of Applied Science

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Real-Time Computing in RTOS, Microprocessor Systems, Sensors & Instrumentation

SKILLS

Software: C, C++, Python, Git VCS, RTOS, Linux, Simulink/MATLAB, CMake, Bash, GDB, ArduPilot, C#, .NET, Java, JavaScript, HTML/CSS, MSSQL, VBA

Technologies & Protocols: ARM (STM32, TI), ESP-IDF, Arduino, Robot Operating System (ROS2), CAN, SPI, I2C, UART, PWM, Ethernet, TCP, UDP, HTTP

Concepts: Robotics, Microcontrollers, Sensors & IMUs, Automation, Controls & Real Time Systems, Signal Processing, IoT (Internet of Things)

EXPERIENCE

Software Engineering Intern

Brampton, ON

MDA Space

September 2025 – Present

- Architected flight software telemetry and command handling pipelines in Java, and YAMCS over TCP, reducing total latency by 30%
- Developed and debugged multi-fidelity robotic arm simulations on Linux clusters for dynamic analysis and 7 DoF kinematic modeling
- Integrated a full-stack GUI with AngularJS, C#, and .NET to interact with simulated robotic manipulation for improving operator training
- Simulated inverse kinematics with Python to minimize error between target and end-effector pose, reducing tested misalignment by >90%
- Built a 7 DoF Canadarm2 digital twin with custom CAD models and control software, enabling self-correction and simulation validation

Embedded Systems Team Lead

Waterloo, ON

Waterloo Space Research Team (Space Soldering Project, Signal Analytics Project)

January 2024 – Present

- Led a team of 4 developers in building a payload for the CANSBX-7 competition, set for autonomous deployment in the stratosphere
- Architected the software system for an SDR-based ML pipeline, defining parameter tuning logic to process transmissions at high altitudes
- Implemented filtering pipelines with SDR communication on GNU Radio to process 20MHz signals for wavelet denoising techniques (DSP)
- Configured the NVIDIA Jetson Linux SoM for autonomous operation with systemd, scripting services in bash to reduce deployment by 40%
- Developed firmware for a microgravity experiment, placing 1st in the CANRGX-7 competition, as awarded by the Canadian Space Agency

Embedded Software Development Intern

Waterloo, ON

AlumaPower Corp.

May 2025 – August 2025

- Built C firmware for STM32, sensors, and distributed controllers, enabling communication and data logging; reducing errors by 55%
- Deployed bridged MQTT communication across edge (Linux) and cloud (Azure) brokers, enabling bidirectional data flow for industrial IoT
- Designed and integrated a CAN-MQTT interface layer on a custom Linux system, enabling reliable messaging with distributed controllers
- Automated board deployment via scripted services and Linux cron jobs for autonomous use, reducing manual boot time by 67%

Digital Systems Engineering Intern

Longueuil, QC

Canadian Space Agency

September 2024 – February 2025

- Leveraged architecture frameworks and tools (TOGAF, SysML) for digital transformation, improving system performance by 15%
- Supported architecture modeling by analyzing system requirements, enhancing digital workflows for space initiatives and missions

Embedded Software Engineering Student

Waterloo, ON

Cooperative & Adaptive Mechatronic Systems (CAMS) Lab

May 2024 – August 2024

- Led the creation of a comprehensive drone system architecture, integrating hardware with ArduPilot for speed, and navigation controls
- Developed a waypoint following ROS2-based control pipeline for autonomous drone operation and precise hover and landing routines

Software Automation Intern

Waterloo, ON

OpenText

January 2024 – April 2024

- Led end-to-end software development and QA testing for an automated workflow application in AppWorks, garnering 30+ internal users
- Conducted Python API and PowerShell script development for optimized data transfers, decreasing manual time spent by 53%

PROJECTS

[ASTRA: Signal Transmission Stratospheric Balloon Payload](#) | C/Python, Signal Processing, Linux FPGAs, HackRF SDR Modules

- Led a team to build the software for a digital signal processing (DSP) payload as part of a Stratospheric Balloon Experiment (CAN-SBX)
- Created SDR interfacing and signal processing scripts for real-time data acquisition and analysis during stratospheric flight operations

[Space Soldering Centrifuge Experiment](#) | C, Nucleo STM32, FreeRTOS, USART Communication, PySerial

- Administered subsystems in C++ and RTOS, driving a 24V motor with an adjustable RPM and collecting data with an accuracy of 96%

[MOSS: GNC Processor-In-The-Loop Validation](#) | C, Nucleo STM32, USART Communication, Hohmann Transfer, Cesium Ion

- Architected a low-cost & scalable framework for rapid prototyping and visualizing GNC algorithms on ARM Cortex based flight controllers